

# Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

---

Албаша Али Омран

2026-03-18

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

## 1. Цели и задачи работы

---

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

## 2. Процесс выполнения лабораторной работы

---

```
alialbasha@alialbasha:~$ ls /etc/ > file.txt
alialbasha@alialbasha:~$ ls >> file.txt
alialbasha@alialbasha:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
cifs-utils
cockpit
colord
containers
```

Рисунок 1: Запись в файл

```
alialbasha@alialbasha:~$ grep .conf file.txt > conf.txt
alialbasha@alialbasha:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
makedumpfile.conf.sample
```

Рисунок 2: Поиск расширения .conf



```
/home/alialbasha/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage03/report/_resources/csl
/home/alialbasha/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage03/report/bib/cite.bib
/home/alialbasha/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage04/report/_resources/csl
/home/alialbasha/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage04/report/bib/cite.bib
/home/alialbasha/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage05/report/_resources/csl
/home/alialbasha/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage05/report/bib/cite.bib
/home/alialbasha/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage06/report/_resources/csl
/home/alialbasha/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project
-personal/stage06/report/bib/cite.bib
/home/alialbasha/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/present
ation/report/_resources/csl
/home/alialbasha/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/present
ation/report/bib/cite.bib
/home/alialbasha/git-extended/.git/hooks/commit-msg.sample
/home/alialbasha/git-extended/.git/objects/f6/cb6ab032d441dfa2076fe2c6160c9968316d2e
/home/alialbasha/git-extended/.git/objects/c5
/home/alialbasha/git-extended/.git/objects/c9
/home/alialbasha/git-extended/.git/objects/cf
/home/alialbasha/git-extended/.git/config
/home/alialbasha/conf.txt
alialbasha@alialbasha:~$
```

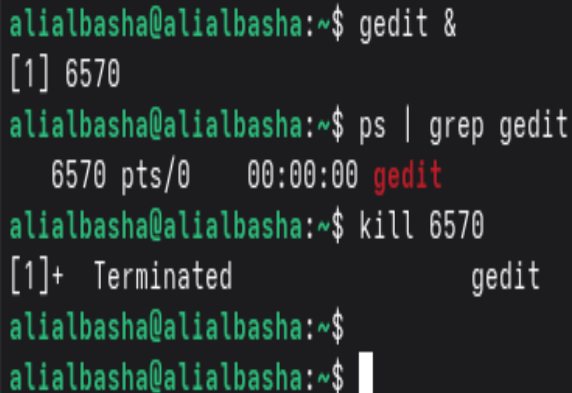
Рисунок 3: Поиск файлов

```
/etc/foomatic/hashes.d/hashes.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
/etc/logrotate.d/httpd
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
/etc/sysconfig/htcacheclean
```

Рисунок 4: Поиск файлов

```
alialbasha@alialbasha:~$  
alialbasha@alialbasha:~$ find ~ -name "log*" > logfile &  
[1] 6531  
alialbasha@alialbasha:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
alialbasha@alialbasha:~$ rm logfile  
alialbasha@alialbasha:~$
```

Рисунок 5: Фоновый запуск процесса



```
alialbasha@alialbasha:~$ gedit &  
[1] 6570  
alialbasha@alialbasha:~$ ps | grep gedit  
6570 pts/0    00:00:00 gedit  
alialbasha@alialbasha:~$ kill 6570  
[1]+  Terminated                  gedit  
alialbasha@alialbasha:~$  
alialbasha@alialbasha:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space
    available on the file system containing each file name argument.  If no file name
    is given, the space available on all currently mounted file systems is shown.
    Space is shown in 1K blocks by default, unless the environment variable
    POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted
    file system, df shows the space available on that file system rather than on the
    file system containing the device node.  This version of df cannot show the space
    available on unmounted file systems, because on most kinds of systems doing so re-
    quires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file
    systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units
        of 1,048,576 bytes; see SIZE format below

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILEs, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size
        is usually smaller, it may be larger due to holes in ('sparse') files, in-
        ternal fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units
        of 1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
        produce a grand total

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
alialbasha@alialbasha:~$ df
Filesystem      1K-blocks      Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 89677192 64843192  59% /
devtmpfs         4011292    0      4011292   0% /dev
tmpfs            4035664    92      4035572   1% /dev/shm
tmpfs            1614268   1952     1612316   1% /run
tmpfs             1024      0        1024    0% /run/credentials/systemd-journald.service
tmpfs            4035668   1608     4034060   1% /tmp
/dev/nvme0n1p3 155186176 89677192 64843192  59% /home
/dev/nvme0n1p2  1992552   518800   1352512   28% /boot
/dev/loop1       115712    115712      0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop2        49280     49280      0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0        68480     68480      0 100% /var/lib/snapd/snap/core24/1349
tmpfs             1024      0        1024    0% /run/credentials/systemd-resolved.service
tmpfs            807132    168     806964   1% /run/user/1031
/dev/sr0         2677920   2677920      0 100% /run/media/alialbasha/Fedora-WS-Live-43
alialbasha@alialbasha:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/4c
4      ./git-extended/.git/objects/4e
4      ./git-extended/.git/objects/c9
4      ./git-extended/.git/objects/3b
4      ./git-extended/.git/objects/1c
4      ./git-extended/.git/objects/cf
4      ./git-extended/.git/objects/9b
4      ./git-extended/.git/objects/5e
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
837796 .
alialbasha@alialbasha:~$
```

Рисунок 10: Запуск команды du



```
/home/alialbasha/git-extended/.git/objects/7c
/home/alialbasha/git-extended/.git/objects/4e
/home/alialbasha/git-extended/.git/objects/c9
/home/alialbasha/git-extended/.git/objects/3b
/home/alialbasha/git-extended/.git/objects/1c
/home/alialbasha/git-extended/.git/objects/cf
/home/alialbasha/git-extended/.git/objects/9b
/home/alialbasha/git-extended/.git/objects/5e
/home/alialbasha/git-extended/.git/logs
/home/alialbasha/git-extended/.git/logs/refs
/home/alialbasha/git-extended/.git/logs/refs/heads
/home/alialbasha/git-extended/.git/logs/refs/remotes
/home/alialbasha/git-extended/.git/logs/refs/remotes/origin
/home/alialbasha/monthly
/home/alialbasha/reports
/home/alialbasha/reports/monthly
/home/alialbasha/reports/monthly/monthly
/home/alialbasha/ski.plases
/home/alialbasha/ski.plases/equipment
/home/alialbasha/ski.plases/plans
/home/alialbasha/australia
/home/alialbasha/play
/home/alialbasha/play/games
/home/alialbasha/play/games/play
alialbasha@alialbasha:~$
```

Рисунок 11: Поиск директорий

### 3. Выводы по проделанной работе

---

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.